

Labor Demand for Digital Skills during Wartime: Evidence from Russia’s Invasion of Ukraine*

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Abstract

This paper examines how wartime disruption shapes the demand for digital skills. We argue that firms partially mitigate war-induced labor shortages and operational frictions by reorganizing production around more digitally intensive tasks and processes. We build a novel dataset covering more than 17 million online job vacancies in Ukraine from 2021 to 2025, using large language models to classify vacancy text and map extracted skills to the official ESCO taxonomy. We link these data to high-frequency measures of attack intensity—ACLED-recorded events and fatalities—and estimate time-series models. Our results show a robust positive association between invasion-related fatalities and the share of postings requiring digital skills, consistent with economy-wide adaptation through digitalization during Russia’s invasion of Ukraine. We further provide evidence that this shift is partly driven by changes in occupational composition, in particular an expansion of military- and security-related roles that are disproportionately digital-intensive, as well as remote work adoption.

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1 Introduction

Armed conflict exerts profound and multifaceted effects on labor markets. It contracts the available workforce through conscription and displacement, while simultaneously heightening the risks associated with physical tasks, leading to acute shortages in both labor supply (Collier and Duponchel 2013) and shifts in the demand for skills (Pham, Talavera, and Wu 2023). Although suggestive evidence indicates that employers alter their skill requirements during wartime, a systematic examination of the mechanisms driving such shifts remains lacking (Pham, Talavera, and Wu 2023).

This paper investigates whether digitalization can mitigate labor shortages and structural disruptions induced by war, invasion, and conflict. We posit that, faced with a reduced labor pool and damaged infrastructure, firms accelerate the adoption of digital technologies, automate routine processes, and reorganize tasks to maintain productivity and operational resilience. International evidence shows that war and conflict lead to the destruction of factories, machinery, and transportation networks, consequently eroding capital stocks (Chauvin and Rohner 2009). Moreover, wars lead to changes in existing institutions and market functioning, raising transaction costs and operational uncertainties (Verwimp, Justino, and Brück 2019). In such an environment, organizations must seek novel operational modalities and technological innovations to survive and adapt (Udomsaph 2023). Evidence of other shocks, such as the COVID-19 pandemic (Forsythe et al. 2020) or recessions (Hershbein and Kahn 2018), have shown how severe disruptions can cause significant changes in skills demand.

We study this question in the context of Russia’s invasion of Ukraine. Conscription and emigration have significantly reduced the domestic labor supply and firms face challenges to fill vacancies (World Bank 2024). In 2023, nearly two out of 10 businesses who had closed reported a lack of workers as the main reason (12 percent in 2024) (World Bank 2024). In addition, the ongoing invasion increases the hazards of on-site work due to the security risk, damaged assets, and increased uncertainty (World Bank 2023; World Bank 2024). Roles requiring on-site attendance—such as plant operator positions—have consistently been hard

to fill across all regions in Ukraine (World Bank 2024).

Both the private and public sector are increasingly digitalizing their services. For example, in December 2023, the Ukrainian government announced e-services for those who have lost their homes due to Russia’s invasion of Ukraine (EU4Digital 2024), online marriage registration services (initially targeted to military personnel) (EAPL 2025), and in July 2024 it launched a one-stop-shop for starting and managing a business online (EGA 2024). The government also provides several e-services targeting IDPs (launched in May 2023) and veterans (launched in July 2025) (Kyiv Global Government Technology Center 2025a). Businesses are accelerating digitalization in response to the invasion, too. A survey on the state of business digitalization in Ukraine conducted in 2024 revealed that six out of 10 surveyed companies have accelerated digitalization (Kyiv Global Government Technology Center 2025b). Both remote work and cost optimization are frequent reasons for this trend among interviewed companies.

Economic theory provides a robust foundation for our hypothesis that wartime pressures accelerate digitalization and reshape labor demand. Capital–labor substitution models predict that scarce labor supplies will prompt firms to substitute toward capital, including software, robotics, and cloud-based solutions (Solow 1956). Task-based frameworks suggest that routine tasks are most susceptible to automation, non-routine cognitive and interactive tasks complement digital tools, and entirely new tasks emerge alongside technological progress (Autor, Levy, and Murnane 2003; Autor 2015; Acemoglu and Restrepo 2018). Moreover, search-and-matching models indicate that prolonged job vacancies and elevated recruitment costs incentivize investments in automation and digital platforms to bridge workforce gaps (Yashiv 2007; Diamond 2011).

By integrating these theoretical perspectives with empirical analysis, this paper delivers the first comprehensive assessment of how armed conflict catalyzes digital transformation in the labor market, with a particular focus on the evolving demand for digital competencies in Ukraine.

To investigate this question, we leverage online job postings collected from an online job vacancy aggregator, the Jooble platform, from January, 2021 to June, 2025. Jooble, a prominent job vacancy aggregator, compiles listings from various Ukrainian job portals, providing a rich dataset for labor market analysis.¹ Utilizing large language models (LLMs), we extract and classify information from online job postings to identify demanded occupations, and required skills. We then map the resulting occupations and skills to the official ESCO classification. The original data also contains regional information, the number of clicks - a proxy for job seekers' interest -, and the date the vacancies were originally posted and expired. Using the official ESCO classification, we calculate the relative demand for digital skills in the Ukrainian labor market.

We combine this data with information on the number of invasion-related events and fatalities provided by ACLED. We then run time-series models to investigate the relationship between Russia's invasion of Ukraine and the relative demand for digital skills. Our results reveal a positive correlation between invasion-related events and fatalities and the demand for digital skills, indicating that war, invasion, and conflict indeed shift the labor market toward digitally intensive tasks and approaches.

The positive association also persists after controlling for seasonal patterns through time fixed effects, broader labor-market conditions captured by weekly vacancy volumes, and the technological shock induced by the rollout of ChatGPT. The results are further robust to alternative time-series specifications, including moving-average smoothing of both the outcome and the regressor, as well as an Almon polynomial lag model that flexibly captures dynamic effects. Controlling for the relative demand for military skills reveals that the invasion increases digital-skill demand at least partly through changes in occupational composition toward military- and defense-adjacent roles, rather than through a pure general trend in digitalization. Remote work adoption emerges as another mechanism linking invasion-related disruption to higher digital-skill requirements.

¹Jooble is a global product-based IT company active in 67 countries and has been operating in Ukraine for over 18 years.

Our study talks to the literature established by Acemoglu and Autor (2011), who discuss the implications of skills, tasks, and technologies on employment and earnings. It is most closely related to a newly emerging literature studying the interaction between demographic change and automation (Acemoglu and Restrepo 2022) and a limited set of studies analyzing the impact of war, invasion, and conflict on innovation/technological change (Anosike (2018), Gross and Sampat (2023), Maltsev (2023), Clemens and Rogers (2023)). Studies that analyze the impact of automation and artificial intelligence on labor demand, as explored by Acemoglu and Restrepo (2019), Acemoglu and Restrepo (2020), and Acemoglu et al. (2021), are also related to our analysis.

Our research offers valuable insights into the dynamic nature of skills demand in Ukraine, informing policymakers, educators, and industry stakeholders in their efforts to develop responsive workforce development strategies. It closes an important gap in labor market intelligence in Ukraine. Accurate labor market information is crucial for efficient labor markets by providing accurate, timely, and accessible data.

2 Data

2.1 Data Extraction and Processing

We construct a large-scale database capturing skills demand in Ukraine, drawing on data provided by Jooble. Jooble is an IT company founded in 2006 that operates Jooble.org, an international job search engine active in 67 countries. The platform functions as an online job vacancy aggregator, using a proprietary algorithm and keyword-based search to scrape Google for job postings across a given country. It also incorporates automated mechanisms to filter out spam, phishing, and inappropriate job listings. The final dataset provided by Jooble is in JSON format and includes information on the job title, job description, region of the job opening, salary details, number of clicks, and the dates the vacancy was created and expired.

In Ukraine, Jooble collects job postings from 3,490 sources—2,570 directly from employers and 920 from other aggregated platforms. Ukrainian employers can register directly on Jooble, enabling the platform to comprehensively cover their job-related websites. The resulting database includes classified job portals, corporate websites, and other platforms featuring job listings. Jooble has been active in the Ukrainian market for 18 years, providing a robust and longitudinal web archive. Its web crawler scans the entire internet every 24 hours to ensure the data remains current.

We process the full Jooble database from January, 2021, to June, 2025, a total set of 17.1 million vacancies. We first clean the dataset by removing duplicates, removing unnecessary text from job titles and vacancy descriptions, and then standardize the language in the job descriptions. We also conduct a language detection exercise given that the original Jooble data for Ukraine comes in multiple languages (Ukrainian, Russian, Czech, Polish, and English). The multilingualism of the job posts is related to the fact that the Jooble website in Ukraine also captures remote jobs, as well as jobs abroad, such as from the Czech Republic, Poland, Germany, as well as several other European countries (Greece, Portugal, Slovakia, and Lithuania). Appendix A provides more details on the data cleaning.

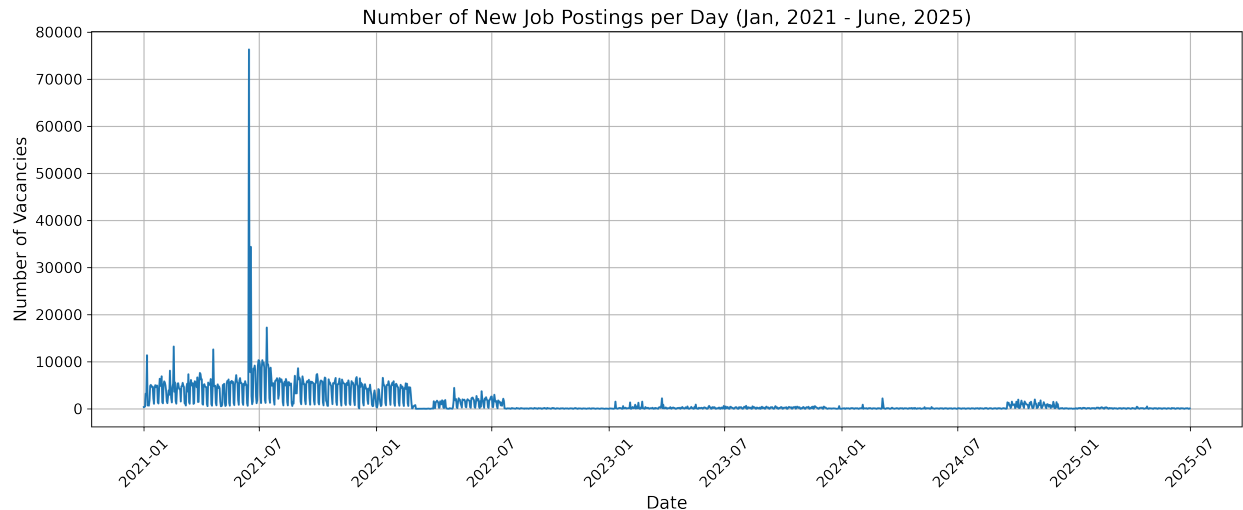
Deduplication Procedure. Given that Jooble aggregates postings from multiple portals, duplication is a central concern. Deduplication in our pipeline is therefore not based solely on record identifiers. In a first step, we apply content-based deduplication by collapsing postings that share identical job titles, descriptions, salary ranges, and regions. This removes repeated appearances of the same underlying vacancy when cross-posted across portals, even if different platform-specific IDs are assigned. In a second step, record identifiers are used to prevent reprocessing of the same vacancy across files and over time. As a result, each underlying vacancy enters the analysis only once, preventing mechanical inflation of skill counts.

We then filter the data to retain only active job postings from Ukraine, yielding a total

of 5.9 million vacancies, as our primary focus is on labor demand within the country. Next, we collapse the data based on the initial date each vacancy was created on the Jooble platform, ensuring that each vacancy appears only once. For the number of clicks—a proxy for job seeker interest or relative attention received by vacancies, which provides insight into potential supply-side preferences—we retain both the average and the maximum number of clicks per vacancy over the total period the vacancy was active. The collapsed data results in 174,640 vacancies posted between January 2023 and June 2025 (when abstracting from ads that were posted to several regions).

Figure 1 depicts the number of newly posted vacancies captured on the Jooble site between January, 2021 and June, 2025. The plot reveals high daily fluctuations in postings throughout 2021 and 2022, with a noticeable drop in frequency and volatility in mid-2022. Since mid-2022, the highest number of vacancies was registered from mid-September 2024 to the beginning of December 2024, suggesting increased scraping activity during that period. These developments demonstrate how scraping activity can confound trends in posting activity.

Figure 1: Number of job vacancies posted per day based on Jooble data



Data Processing Overview. The analytic dataset is constructed through a structured multi-stage pipeline. Raw job vacancy postings are collected from Jooble in JSON format and subjected to cleaning, deduplication, and language filtering. In the next step, candidate skills are extracted from vacancy texts using multilingual sentence embeddings and matched to the ESCO skill taxonomy based on semantic similarity. Vacancies are then classified into ESCO occupations using a large language model and enriched with occupation-level ESCO skills to capture both explicitly stated and implicitly related requirements. Geographic information is standardized and harmonized, after which processed data is aggregated into weekly and monthly panels. Finally, these panels are merged with ACLED measures of invasion-related events and fatalities to construct the final analytic sample used in the empirical analysis. A detailed technical description of each processing stage is provided in Appendix A.

Modularity and Extensibility. The pipeline is designed as a modular and extensible framework. While the present paper focuses on empirical analysis, the underlying architecture separates data ingestion, skill extraction, classification, enrichment, and aggregation into configurable stages. This design allows future extensions—such as alternative taxonomies, updated models, or additional outcome measures—to be incorporated through metadata or configuration changes without restructuring the core workflow.

2.2 Skills and Occupation Extraction

We next apply large language models to extract unified skills and occupations from our cleaned online job vacancy data. We use the *paraphrase-multilingual-mpnet-base-v2* model from *Sentence-Transformers* to generate dense vector embeddings of job vacancy texts. *Sentence-Transformers* is a framework built on top of pretrained transformer models (such as BERT, RoBERTa, or MPNet) that enables efficient computation of sentence and paragraph embeddings suitable for semantic similarity, clustering, and information retrieval tasks. The model supports over 50 languages and captures semantic similarity across sentences, enabling

multilingual classification and clustering of skill requirements. It is based on MPNet (Masked and Permuted Pre-training for Language Understanding), a transformer architecture that combines the strengths of BERT and XLNet by using both masked and permuted language modeling, resulting in improved performance on semantic understanding tasks.

We then compare the vector embeddings of job vacancy texts with the complete list of official ESCO skills (both the preferred label and alternative labels), which are likewise encoded as vector embeddings. Using semantic similarity², we apply different cosine similarity thresholds at distinct stages of the pipeline. In the initial candidate skill extraction step, we use a cosine similarity threshold of 0.5 to maximize recall when matching vacancy texts to ESCO skill labels. In subsequent occupation classification and skill enrichment steps, a stricter threshold of 0.8 is applied, where greater precision is required when aligning job titles and occupations with the ESCO taxonomy. This approach allows us to extract up to 25 skills per job vacancy. Based on the list of skills and online job vacancy description, we then classify each job vacancy into ESCO occupations using *gpt-4o-mini* and *gpt-4.1-mini*. We then merge the full list of ESCO skills related to each ESCO occupation to the final job vacancy database. Lastly, we harmonize the regional information using *OpenAI’s GPT-4o API*. Appendix A provides more details on the skills and occupation extraction process.

To account for differences in scraping activity, we analyze the relative demand for digital skills (not the absolute one). We construct this measure by counting all digital skills, as identified by the official ESCO classification, and dividing it by the total number of skills for a certain period of time (e.g., at the weekly or monthly level).

2.3 Data Limitations

Importantly, the data may not fully reflect the true structure of labor demand across Ukraine. Since Jooble aggregates postings primarily from online sources, occupations that are typically filled through informal networks, in-person recruitment, or public employment centers—such

²Semantic similarity refers to how closely two texts align in meaning, typically measured using cosine similarity between their vector embeddings in a shared semantic space.

as public sector jobs, rural work, or highly specialized roles—may be underrepresented. Additionally, disparities in internet access, especially in invasion-affected or occupied regions, may further bias the coverage of job ads, limiting the representativeness of the data across all oblasts and sectors.

Moreover, the prominence of certain skills—particularly generic ones like communication or teamwork—may reflect common phrasing in job ads rather than actual labor market needs. Additionally, the use of semantic matching for skill extraction depends on the quality of job descriptions and the ESCO taxonomy’s coverage, which may miss specialized or emerging skills. We conduct robustness tests and random manual checks to validate the quality of our classifier approach. Appendix A outlines the details.

One important limitation of the data is that observed trends in job postings may partly reflect fluctuations in web scraping activity rather than genuine shifts in labor market demand. Variations in scraping intensity—due to technical issues, platform changes, or incomplete extractions—can introduce noise into the data. To mitigate this, we analyze changes over time over larger time periods (a 6-month time period) and normalize occupation- or skills-specific counts by the total number of job ads posted per day, thereby focusing on the relative share of postings by occupation or skill.

Overall, while useful, the Jooble data should be interpreted with caution and ideally complemented by other sources to gain a more comprehensive view of skill demand in Ukraine.

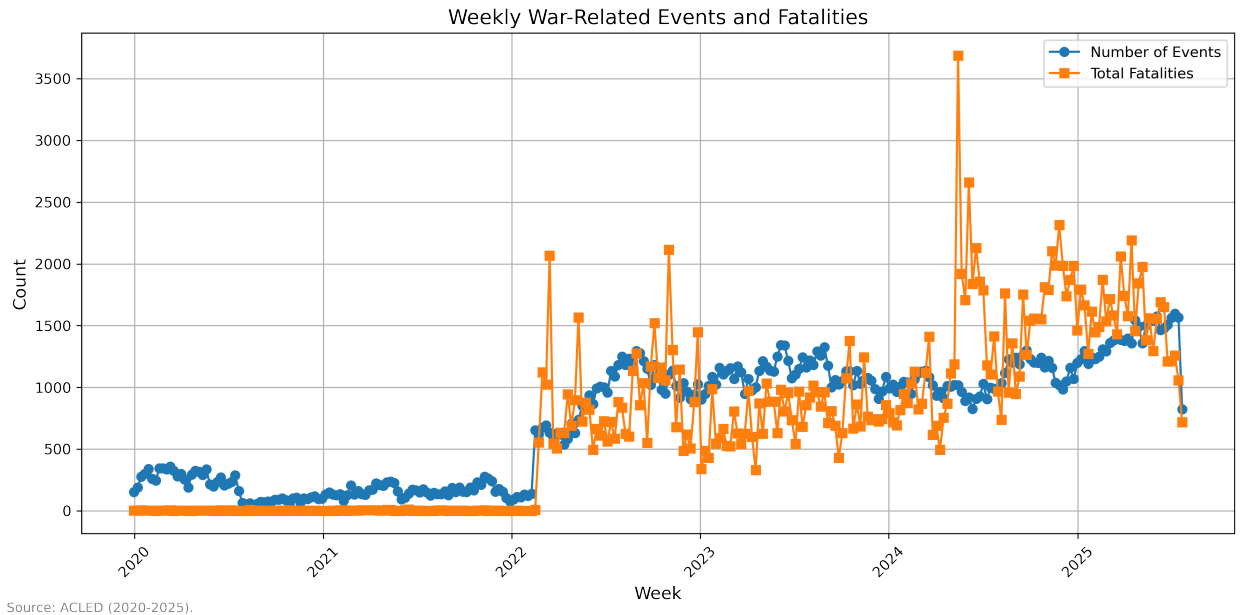
2.4 ACLED data

To analyze how Russia’s invasion of Ukraine has affected the demand for digital skills in Ukraine, we rely on data from the Armed Conflict Location and Event Data Project (ACLED) (ACLED 2023). ACLED offers a high-resolution, event-level dataset capturing instances of political violence, protests, and strategic developments across more than 200 countries and territories. Each record represents a discrete incident with standardized fields for event type (e.g., battle, riot, demonstration), date, precise location (down to city or vil-

lage), and actors involved (including state forces, rebel groups, militias, and civilians). The dataset also records reported fatalities, enabling quantitative assessment of attack intensity and human impact over time. ACLED compiles data through a multi-stage process combining automated media scraping, human coding, and cross-validation by regional experts.³

Figure 2 depicts the total weekly number of invasion-related events and fatalities in Ukraine. The figure clearly reveals how the full-scale invasion by Russia resulted in a significant increase in the number of violent events and fatalities in Ukraine.

Figure 2: Total weekly number of invasion-related events and fatalities in Ukraine (2020 - 2025)



3 Methodology

To shed further light on the impact of Russia’s invasion of Ukraine on the demand for digital skills we run the following regression:

³Source materials include local and international news outlets, government statements, and verified social media reports, which are triangulated to maximize accuracy and minimize reporting bias. Rigorous quality controls detect duplicate or false reports and harmonize actor nomenclature across geographies.

$$Y_t = \beta_0 + \beta_1 \log(\text{fatalities})_t + \sum_{m=1}^{11} \delta_m D_{m,t} + \varepsilon_t \quad (1)$$

where

- Y_t is the average share of digital skills in period t ,
- $\log(\text{fatalities})_t$ is the total number of fatalities per 1,000 in time period t ,
- δ_m captures time period fixed effects,
- β_0 is the intercept, β_1 measures the effect of fatalities on the digital skill share, and
- ε_t is an iid error term.

$D_{m,t} = 1$ if time period t is calendar quarter or month m (with one quarter or month omitted as the reference) and 0 otherwise. These dummy variables absorb any recurring seasonal or calendar-month shocks—holidays, fiscal cycles, training programs—so that β_1 reflects the within-month/quarter relationship between fatalities and digital skill share. By effectively comparing each time period to itself across years, this specification purges calendar-driven noise and sharpens the precision and validity of our key coefficient.

We run regressions at the weekly and monthly level. We estimate our main models at the weekly frequency, which captures high-frequency adjustments in hiring in response to shocks induced by Russia’s full-scale invasion of Ukraine. Weekly variation is economically meaningful in Ukraine, where missile strikes, power outages, and front-line movements occur on a weekly rhythm. To ensure that results are not driven by short-term volatility, we re-estimate the specifications at the monthly level.

To account for the highly skewed distribution of attack intensity and to obtain economically meaningful elasticities, we model fatalities (and events) in logarithms. Weekly conflict indicators exhibit substantial variation and frequent extreme spikes, making a linear specification inappropriate and highly sensitive to outliers. The log transformation compresses

these extreme values, stabilizes variance, and improves the linear fit between conflict intensity and labor-demand responses. Substantively, the log specification captures the idea of diminishing marginal effects—an increase from 0 to 10 events is likely to have a much larger impact on employer behavior than an increase from 100 to 110. Interpreting the coefficient as a semi-elasticity also provides clearer economic meaning, linking percentage changes in conflict intensity to percentage-point changes in digital skill demand.

We also estimate a polynomial distributed lag model (Almon lag), which imposes a smooth functional form on the lag coefficients of conflict intensity. This approach reduces multicollinearity across adjacent weekly lags, improves precision, and yields a smooth dynamic response function. The specification is well-suited to our setting, where the effect of conflict intensity is expected to evolve gradually rather than discontinuously week by week.

4 Labor Demand for Digital Skills

Figure 3 reveals that the relative demand for digital skills has increased over the period January, 2023, to June, 2025. Plotting the monthly average share of digital skills demanded against the monthly number of fatalities registered in Ukraine indicates a positive correlation between both variables (figure 4). Both the Pearson correlation and Spearman correlation confirm the positive relationship (0.527 and 0.481 respectively).

Figure 3: Share of digital skills demand in all skills based on Jooble data (Jan, 2021-June, 2025)

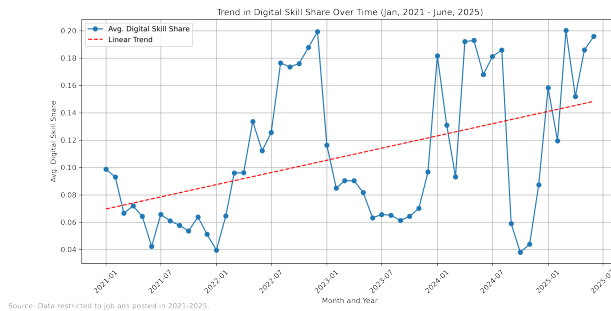


Figure 4: Scatter plot of monthly fatalities and average share of digital skills demand in all skills

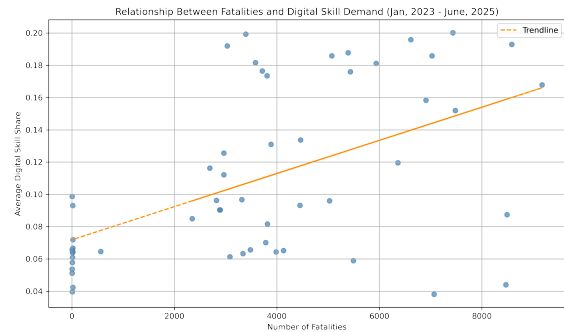


Table 1: OLS Regression Results

	(1)	(2)	(3)	(4)
Log(Fatalities)	1.189*** (0.092)	1.198*** (0.147)	1.425*** (0.326)	1.203*** (0.147)
Quarter FE	No	Yes	Yes	No
Year FE	No	No	Yes	No
Month FE	No	No	No	Yes
Observations	235	235	235	235

Notes: Standard errors in parentheses. Model (1) uses HC3 SE. Models (2)–(4) use HAC SE with 4 lags. Quarter, year, and month fixed effects included as indicated.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: OLS Regression Results

	(1)	(2)	(3)	(4)
Log(Events)	3.373*** (0.276)	3.427*** (0.444)	4.719*** (0.842)	3.495*** (0.431)
Quarter FE	No	Yes	Yes	No
Year FE	No	No	Yes	No
Month FE	No	No	No	Yes
Observations	235	235	235	235

Notes: Standard errors in parentheses. Model (1) uses HC3 standard errors. Models (2)–(4) use HAC standard errors with 4 lags. Quarter, year, and month fixed effects included as indicated.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Robustness

To ensure that our findings are not driven by high-frequency noise in the weekly data, we re-estimate our models using monthly aggregates of digital-skill demand and conflict intensity. Monthly aggregation reduces week-to-week volatility in both hiring behavior and attack patterns, allowing us to test whether the underlying relationship persists at a more stable temporal resolution. The monthly regressions confirm a strong and statistically significant

relationship between conflict intensity and the demand for digital skills (see table B1). Across specifications, a one-unit increase in the log of monthly events or fatalities predicts between 1.0 and 5.5 percentage points higher digital-skill intensity in posted vacancies. These effect sizes are substantially larger than in the weekly models, where comparable elasticities typically range from 0.8 to 1.3 percentage points. This divergence is consistent with aggregation smoothing out short-run noise: monthly data reduce high-frequency volatility arising from week-to-week operational adjustments, irregular attack timing, and reporting spikes. As a result, the signal-to-noise ratio improves, and the estimated coefficients capture more persistent adjustments in firms’ hiring strategies. In other words, the monthly specifications more clearly isolate the underlying structural response of employers—namely, a systematic shift toward digitally intensive work—whereas weekly regressions reflect both this trend and transient fluctuations.

To examine whether the documented relationship between attack intensity and digital-skill demand is driven by changes in the overall volume of online job postings, we augment the baseline model with weekly vacancy counts. This variable behaves like a control for aggregate labor demand — it helps isolate the invasion-related effect. Column 1 in table 3 depict the results. The results show that the coefficient on log fatalities remains positive and statistically significant ($\beta = 0.74$, $p = 0.003$), although its magnitude declines relative to the baseline, suggesting that part of the effect is mechanically related to contractions in hiring activity during high-attack weeks. Consistent with this interpretation, vacancy volume has a negative and significant association with digital-skill share: when hiring declines—which typically affects non-digital occupations more strongly—the relative prevalence of digital-intensive vacancies rises. Overall, the main effect remains robust to controlling for labor-market volume effects. The pattern of results holds when using the number of invasion-related events as the main explanatory variable (see Column 1 in table 4).

To test whether the invasion–digital skill relationship is confounded by broader technological shifts, we augment the baseline specification with an indicator for the general rollout

of ChatGPT. Column 2 in table 3 depict the results. The coefficient on log fatalities remains large and highly significant ($\beta = 1.55$, $p < 0.001$), suggesting that the rise in digital-skill demand during periods of intensified attacks is not driven by concurrent advances in AI technologies. The ChatGPT rollout variable enters with a negative sign ($\beta = -2.65$, $p = 0.067$), indicating a modest reduction in digital-skill intensity following the release of generative AI tools, though the estimate is imprecisely measured. This pattern of results could mean that ChatGPT substituted for some digital-skill requirements: employers now require fewer baseline digital skills because AI handles tasks that previously needed human digital proficiency. Importantly, including this control does not attenuate the main effect of attack intensity, reinforcing the robustness of the core relationship. The findings are robust to using the number of invasion-related events as the main explanatory variable (see Column 2 in table 4).

Table 3: OLS Regression Results with Additional Controls

	(1) Vacancies	(2) ChatGPT	(3) Military Share	(4) All Controls
Log(Fatalities)	0.744** (0.254)	1.546*** (0.199)	1.022*** (0.186)	0.827*** (0.306)
Vacancy Count	-9.55e-05** (4.39e-05)			-1.00e-04** (5.68e-05)
ChatGPT Rollout		-2.653* (1.447)		-3.942*** (1.343)
Military Skill Share			96.80 (80.34)	157.70** (73.55)
Quarter FE	Yes	Yes	Yes	Yes
Observations	235	235	235	235
R-squared	0.322	0.328	0.316	0.390

Notes: Standard errors in parentheses. All models estimated using HAC standard errors with 4 lags. Dependent variable: weekly digital-skill share (percentage points). Quarter fixed effects included in all specifications.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To assess the robustness of the relationship between attack intensity and the demand for digital skills, we also estimate specifications that rely on moving-average smoothing of the underlying weekly series. Because weekly fatalities and vacancy skill shares exhibit

Table 4: OLS Regression Results with Additional Controls (Event Intensity)

	(1) Vacancies	(2) ChatGPT	(3) Military Share	(4) All Controls
Log(Events)	1.742* (0.888)	4.652*** (0.621)	2.827*** (0.594)	2.330** (1.014)
Vacancy Count	-1.00e-04** (5.63e-05)			-1.00e-04** (6.47e-05)
ChatGPT Rollout		-2.964* (1.558)		-4.192*** (1.422)
Military Skill Share			114.20 (82.24)	171.44** (74.62)
Quarter FE	Yes	Yes	Yes	Yes
Observations	235	235	235	235
R-squared	0.307	0.307	0.299	0.384

Notes: Standard errors in parentheses. All models estimated using HAC standard errors with 4 lags. Dependent variable: weekly digital-skill share (percentage points). Quarter fixed effects included in all specifications.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

considerable short-term volatility, we replace the raw values with 4- and 8-week moving averages and re-estimate the baseline model using these smoother inputs. This approach reduces the influence of idiosyncratic weekly spikes and allows us to evaluate whether the estimated relationship is driven by short-run noise or reflects broader medium-run dynamics. The results remain strong and statistically significant when both the explanatory variable and the outcome are smoothed, indicating that the positive association between attack intensity and digital-skill demand is not an artifact of high-frequency fluctuations but persists when using multi-week aggregates.

The results indicate no contemporaneous effect of fatalities on skill requirements, but a strong and statistically significant impact emerges with a two-week delay. The coefficient on fatalities at lag 2 is 0.81 percentage points ($p = 0.029$), while the effects at lags 0, 1, and 3 are small and statistically insignificant (figure 5). Summing across the four lags yields a cumulative effect of 1.28 percentage points per log unit increase in fatalities ($p < 0.001$), equivalent to approximately 0.9 percentage points for a doubling of fatalities. This dynamic pattern suggests that firms adjust digital-skill requirements with a short lag, consistent with

operational posting timelines.

While moving averages smooth the series themselves, they do not reveal how the timing of invasion shocks affects employer behavior. To examine the dynamic response more explicitly, we estimate a polynomial distributed-lag (Almon) model, which imposes a smooth functional form on the lag coefficients of weekly fatalities. Instead of collapsing multiple weeks of attack intensity into a single moving-average measure, the Almon specification retains the full lag structure but constrains adjacent lag coefficients to lie on a smooth polynomial curve. This approach mitigates multicollinearity across highly correlated weekly lags, improves statistical precision, and yields an interpretable lag profile showing how the effect builds, peaks, and fades over time. In our setting, the polynomial lag model confirms a delayed response: the impact of attack intensity on digital-skill demand strengthens over the first two weeks and remains positive in the cumulative effect, fully consistent with—yet more informative than—the moving-average robustness checks.

The implied lag coefficients reveal no contemporaneous response of digital-skill demand to increases in attack intensity, but a strong positive effect emerges with a one- to two-week delay (figure 6). The impact peaks at lag 2 (0.66 percentage points per log unit of fatalities) and tapers off by week 3, consistent with firms needing time to adjust job postings. The cumulative effect over the full 0–3 week window is 1.28 percentage points, which closely matches our baseline and moving-average specifications. These results confirm that the relationship between attack intensity and digital-skill demand is robust and operates through short-run dynamic adjustments rather than immediate reactions.

The distributed-lag and polynomial-lag Almon specifications yield a highly consistent picture of the dynamic response of digital-skill demand to conflict intensity. In the unrestricted distributed-lag model, the effect of fatalities is negligible contemporaneously but becomes sizeable and statistically significant after two weeks, with a cumulative impact of 1.28 percentage points over the subsequent month (approximately 0.9 percentage points for a doubling of fatalities). The Almon polynomial lag model, which smooths the lag coef-

Figure 5: Coefficient plot - moving-average smoothing specification

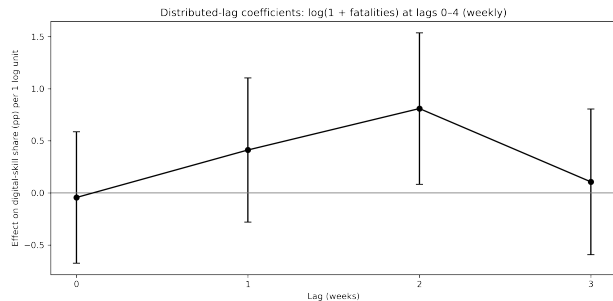
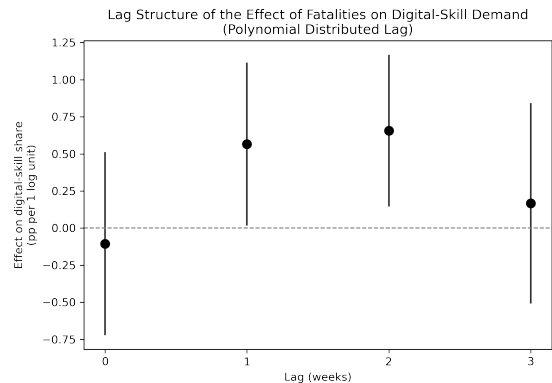


Figure 6: Coefficient plot - Almon specification



ficients to reduce multicollinearity and impose a plausible dynamic structure, recovers an almost identical lag profile: no immediate effect in week 0, a sharp increase in weeks 1–2, and a tapering by week 3. The peak effects in both models occur at a two-week delay, and the cumulative impacts are nearly identical. Taken together, these results indicate that the relationship is not driven by a single noisy weekly coefficient but reflects a stable and interpretable dynamic adjustment: firms increase digital-skill requirements shortly after periods of heightened invasion-related activities, with effects materializing over a two- to three-week window.

6 Mechanisms

A natural concern is that the positive relationship between attack intensity and digital-skill demand may simply reflect compositional shifts in vacancies—for example, an increase in military-oriented job postings that require higher digital literacy. To test this channel, we augment the baseline model with the average military-skill share of vacancies in each week. Column 3 and 4 in table 3 show the coefficients. While the results show no evidence for this mechanism when abstracting from additional controls (Column 3), it turns significant when controlling for aggregate labor demand and the rollout of ChatGPT. Independently of

the specification, the coefficient on log fatalities remains large and highly significant (1.02 percentage points per log unit, $p < 0.001$). This pattern is consistent with a mechanism in which intensified attacks shift hiring toward defense-adjacent occupations that are disproportionately digital-intensive. In other words, once aggregate labor demand and technological trends are netted out, the military channel emerges as a key pathway linking conflict to the rise in digital-skill requirements. Similar findings emerge when using the number of invasion-related events as the main explanatory variable (see Column 3-4 in table 4).

We next investigate if Russia’s invasion of Ukraine has led to a shift to remote jobs. Including the weekly share of vacancies allowing for remote work reveals that remote work is a dominant pathway linking attack intensity to digital-skill demand. Table 5 depicts the results. The remote share variable is strongly and precisely estimated (approximately 35 percentage points, $p < 0.001$), and once it is included, the direct effect of fatalities and invasion-related events attenuates substantially, indicating that a meaningful portion of the increase in digital-skill requirements is mediated by a shift toward remote, digitally enabled work arrangements since 2022.

Table 5: OLS Regression Results: Remote Work as a Mechanism

	(1) Remote + Fatalities	(2) Remote + Events
Log(Fatalities)	0.101 (0.127)	
Log(Events)		0.754* (0.402)
Remote Share	34.676*** (1.424)	34.450*** (1.348)
Vacancy Count	-4.30e-05** (1.72e-05)	-2.90e-05** (1.40e-05)
ChatGPT Rollout	-4.098*** (0.463)	-4.385*** (0.469)
Quarter FE	Yes	Yes
Observations	235	235
R-squared	0.876	0.878

Notes: Standard errors in parentheses. Both models estimated using HAC standard errors with 4 lags. Dependent variable: weekly digital-skill share (percentage points).

Column (1) controls for invasion-related fatalities; Column (2) uses ACLED event counts. The coefficient on remote work reflects the weekly share of postings explicitly mentioning remote or distance work in English, Ukrainian, or Russian.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Conclusion

This paper examines how wartime disruption shapes the demand for digital skills. Using a dataset of more than 17 million online vacancies in Ukraine—classified with large language models and merged to ACLED measures of invasion-related intensity—we document a robust positive association between weekly fatalities and the share of postings requiring digital skills. The results hold when accounting for multiple forms of seasonality, applying moving-average smoothing, estimating Almon polynomial lag specifications, and controlling for broader labor-market conditions, including aggregate vacancy volume and the roll-out of ChatGPT. We further show that part of the increase in digital-skill demand is linked to shifts in occupational composition, particularly a rise in military- and defense-related hiring, as well as a shift to remote work.

Taken together, the results suggest that firms respond to invasion-induced labor scarcity, asset damage, and uncertainty by shifting toward digitally intensive production and coordination, leading to increased labor demand for digital skills. These effects seem to be transmitted through economy-wide channels rather than purely local shocks.

Based on these results, investments in resilient digital infrastructure that underpins remote operations during shocks is recommended. Similarly, policymakers could roll-out interventions that support firm adoption of digital tools. Moreover, policymakers should prioritize job-relevant digital training, with targeting to displaced workers and sectors facing acute labor shortages.

Our estimates are subject to important caveats. First, our findings are associative, not causal, and rely on online vacancy data that may under-represent the public and informal sectors and urban–platform biases. Additionally, LLM classification, while scalable, is not error-free and subject to limitations.

In conclusion, the evidence indicates that war, invasion, and conflict catalyzes demand for digital capabilities. Strengthening digital infrastructure and human capital can help sustain employment and productivity during war, invasion and conflict and accelerate recovery once conditions stabilize.

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Online Appendix

A Data Appendix

Dataset Overview

The dataset analyzed in this study comprises job vacancy advertisements collected by Jooble, a job search engine founded in 2006 and active in 67 countries. For Ukraine, Jooble collects postings from 3,490 sources, including 2,570 direct employer feeds and 920 from third-party aggregators. The collected data spans from January 2021 to June 2025 and is provided in JSON format. Each record represents a unique job listing and includes structured fields (e.g., job title, region, salary, number of clicks) and unstructured text (e.g., job descriptions).

Key variables include:

- **id**: Unique identifier of the job advertisement.
- **title, description**: Raw job title and full text description.
- **region**: Geographical location of the job.
- **min/max salary, currency, salary rate**: Compensation details (often missing or inconsistent).
- **date created, date expired**: Posting start and end dates.
- **number of clicks**: Proxy for job ad popularity.

Initial analysis shows that job descriptions are predominantly in Ukrainian, though some ads use Russian, English, Polish, or Czech. Salary information is inconsistently reported. Duplicate listings are frequent due to repeated postings or slight changes in metadata. Despite these limitations, the dataset offers rich insight into real-time labor demand trends across regions and occupations.

Pipeline Overview

We developed a modular data processing pipeline using Python and Jupyter Notebooks to clean, enrich, and classify the job ads for labor market analysis. The pipeline includes the following core stages:

1. **Initial Preprocessing**
2. **Candidate Skills Extraction**
3. **ESCO Occupation Classification**
4. **ESCO Skill Enrichment**
5. **Region Enrichment**
6. **Final Data Consolidation**

Each stage was implemented with traceable outputs and a status tracking system to ensure reproducibility and error handling.

Key Processing Steps

Initial Preprocessing (Stage 1) Raw JSON files were parsed, deduplicated (based on ID), and cleaned. Text fields were sanitized using regex functions to remove dates, salary mentions, URLs, emojis, and special characters. Language detection was performed using `fast-langdetect`, and ads were retained only if both title and description were in Ukrainian, Russian, English, Polish, or Czech. Cleaned files were saved as binary `.pkl` for efficiency.

Multi-step Deduplication. Duplication is a critical issue in aggregated vacancy data, particularly when postings are sourced from multiple portals. Deduplication is therefore implemented in multiple stages.

First, we apply content-based deduplication by collapsing postings that share identical job titles, descriptions, salary ranges, and regions. This procedure removes repeated appearances

of the same underlying vacancy when it is sourced from multiple portals or aggregators, even if distinct record identifiers are assigned.

Second, we enforce identifier-based deduplication across processing runs to ensure that the same vacancy is not reintroduced when files are processed incrementally over time. This safeguards against mechanical duplication arising from batch updates or partial reprocessing.

As a result, each vacancy enters the analytical dataset only once.

Multi-Region Postings. Identical job advertisements may legitimately appear across multiple regions (e.g., nationwide or multi-location postings). This represents a distinct issue from cross-portal duplication. During intermediate processing stages, such postings are temporarily aggregated to optimize computational efficiency and avoid redundant downstream processing. Importantly, regional information is not discarded. In the final analytical stage, vacancies are re-expanded and assigned to regions, ensuring that regional variation is fully preserved.

Skill Translation (Interim Step) Since the ESCO framework does not natively support Russian, we translated 13,700 English ESCO skills into Russian using the OpenAI GPT-4o API. This enabled effective skill extraction from Russian-language job ads.

Candidate Skills Extraction (Stage 2) Using *sentence-transformers* and the *paraphrase-multilingual-mpnet-base-v2* model, we embed sentences from job descriptions and compare them to pre-encoded ESCO skill vectors. Matches are selected via cosine similarity with a threshold of 0.5, chosen to prioritize recall at the candidate extraction stage. This relatively permissive threshold ensures that potentially relevant skills are not prematurely excluded in the initial filtering step. Up to 25 top-matching skills are stored per job ad. In parallel, the same embedding model is used to classify each vacancy’s work modality (remote, in-office, or hybrid) by comparing its text to a small set of reference embeddings. This addition augments the dataset with a structured job-type indicator while leaving the core skill-extraction

process unchanged.

Occupation Classification (Stage 3) If a job ad contains at least five extracted skills, these skills are used to predict its ESCO occupation using a lightweight model (e.g., gpt-4o-mini) in batch mode; otherwise, a truncated job description (up to 1,000 characters) is used as fallback input. After this initial pass, all low-confidence or likely misclassified cases undergo a second classification step using a more capable model (gpt-4o), which provides higher accuracy for ambiguous or noisy titles. Prompt-engineered responses return structured ESCO codes and titles, primarily at the 4-digit level, with occasional 5-digit outputs.

ESCO Skill Enrichment (Stage 4) In Stage 4, the ESCO skill enrichment process builds on the outputs from Stage 3 by linking each classified job title to the official ESCO taxonomy. The code first attempts to match cleaned job titles directly to ESCO’s preferred occupation labels, and if that fails, it checks alternative labels. Remaining unmatched cases are resolved through fuzzy matching against both preferred and alternative ESCO labels using a stricter similarity threshold of 0.8. This higher threshold prioritizes precision at the enrichment stage, where accurate alignment between job titles and official ESCO occupations is critical. However, despite this enrichment, around 30 percent of occupations remain unclassified, reflecting limitations of fuzzy matching and the challenges of aligning noisy job titles with the ESCO taxonomy. Moreover, a 0.8 similarity score still results in misclassifications under small deviations in the wording. For example, "location manager" gets misclassified as "education manager" because the language is close enough to generate a high similarity score despite the classification being different.

Once an occupation is identified, its ESCO identifiers and ISCO codes are added, and the vacancy is enriched with all official skills associated with that occupation. This way, even skills not explicitly mentioned in job postings are captured, ensuring a more complete representation of labor market requirements.

Region Enrichment (Stage 5) Original region strings were parsed and standardized into English city, district, region, and country using GPT-4o. Where possible, geographic coordinates (latitude, longitude) were appended. The enriched region data enables spatial analysis and mapping.

Final Data Consolidation (Stage 6) Outputs from all stages were merged using the job ID. Monthly datasets were generated for efficient access and export. The final dataset includes fields for cleaned text, detected language, extracted and ESCO skills, occupation codes, and enriched geographic metadata.

Output Structure and Use Cases

The enriched dataset is stored in JSON format and includes the following categories:

- **Job metadata:** id, title, description, date created/expired, clicks
- **Cleaned text:** clean title, clean desc, detected languages
- **Extracted skills:** skill URIs and human-readable labels
- **ESCO classification:** occupation codes and titles, ESCO-derived skills
- **Geography:** original and standardized location, coordinates

This comprehensive structure supports temporal trend analysis, skill gap estimation, regional comparison, and demand forecasting. Exported monthly files also enable integration into dashboards, APIs, and modeling systems.

Reproducibility and Pipeline Documentation. To support transparency and reproducibility, we are preparing a dedicated reproducibility package that documents the full data-processing pipeline, including repository structure, modular workflow stages, software

versions, and environment dependencies. The package will provide clear guidance on environment setup, required libraries, version control, and execution scripts to facilitate replication and collaborative use across teams and institutions.

The reproducibility archive is intended to be made publicly available and will complement the methodological description provided in this appendix by enabling interested researchers and partners to replicate, audit, and extend the analysis in a structured and transparent manner.

The package will also include example configuration files and instructions for generating the final analytic dataset used in the empirical estimations.

Verification, Logging, and Operational Stability. Given the computational intensity of several pipeline stages, the workflow incorporates lightweight verification and logging mechanisms to ensure operational stability. Automated checks include row-count validation, uniqueness constraints on identifiers, and summary statistics of missing values to detect unexpected changes in input data across processing stages. Where applicable, hash comparisons are used to monitor potential alterations in intermediate files.

Processing is tracked at the file level with stage-level logging that records execution status and the creation of output artifacts. Because individual stages can be time-consuming, the pipeline is designed to resume from the last successfully completed file in the event of interruption or failure, rather than restarting the entire workflow. This restart mechanism improves robustness to partial runs and facilitates stable, collaborative use across environments.

Framework Orientation and Future Deployment. The pipeline described in this study is structured as a research-oriented reference implementation rather than a deployment manual. Its modular architecture allows advanced quantities, alternative classifications, or additional analytical layers to be integrated progressively as needed.

By separating configuration parameters (e.g., similarity thresholds, model selection, ag-

gregation levels) from core processing logic, the framework preserves flexibility for maintenance, adaptation, and potential operational deployment. Should live or continuous analysis become relevant, incremental adjustments can be implemented through metadata or configuration updates without altering the conceptual structure documented in this paper.

B Robustness Checks

Table B1: Monthly OLS Regression Results

	(1) Events	(2) Events	(3) Fatalities	(4) Fatalities
Log(Events)	3.154*** (0.707)	5.542*** (1.248)	1.041*** (0.222)	1.536*** (0.414)
Quarter FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes
Observations	54	54	54	54
R-squared	0.294	0.555	0.299	0.518

Notes: Standard errors in parentheses. All models use HAC standard errors with 4 lags. Dependent variable: monthly digital-skill share (percentage points). Panel (1) includes quarterly fixed effects; Panel (2) includes both quarterly and year fixed effects; Panel (3) is a simplified model with only fatalities and quarter FE; Panel (4) adds year FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B1: Top 50 in-demand skills based on Jooble data (Jan, 2021 - May, 2025)

